PCCOE INTERNATIONAL GRAND CHALLENGE: 2026 | PUNE, INDIA

THEME: AI FOR CLIMATE CHANGE

FARMSHIELD AI

AI-Powered Multi-Hazard Climate Impact & Early Intervention System for Agriculture

"FarmShield converts a weather warning into a crop-specific intervention, then uses field evidence to update the risk."

"From Climate Prediction to Crop Protection"



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PROBLEM STATEMENT/IDEA

Problem Statement/Idea

Climate risk is no longer a single isolated forecast — compounding shocks disrupt specific crop cycles.

⚡ THE VULNERABILITY CHAIN: Weather Event ➔ Unnoticed Crop Stress ➔ Preventable Crop Loss

Excess Rainfall
Waterlogging

Heat Stress
Flower / Fruit Stress

Low Rainfall
Water Stress

Temperature Shock
Growth Disruption

"Farmers get weather forecasts.
They need crop-specific actions."

PROBLEM STATEMENT/IDEA

Problem Statement/Idea: The Core Gap

Weather Forecast ≠ Crop Protection — Why general meteorological forecasts fail to stop crop loss.

STATUS QUO: WEATHER APPS

Typical Flow:

Weather Satellite ➔ 'Rain Expected: 70%' ➔ Confused Farmer

❌ **Tells what the sky does, not what the crop needs**
A rain forecast does not explain field risk.

❌ **Blind to growth stages**
30mm rain is good for sugarcane, but fatal for flowering peas.

❌ **Zero field verification**
One-way broadcast with no check on whether crops survived.

FARMSHIELD AI: CLOSED LOOP

FarmShield Flow:

Weather ➔ Hazard ➔ Crop Stage ➔ Action ➔ Photo ➔ Verify

✓ **Answers: 'What does this weather mean for THIS crop?'**
Converts climate volatility into localized crop vulnerability.

✓ **Growth-stage specific rules**
Dynamic thresholds for germination, flowering, podding, and harvest.

✓ **Field verification loop**
Farmer photo upload allows AI to verify crop stress trend over time.

THE DIFFERENCE: Normal weather apps predict weather. FarmShield predicts CROP RISK and verifies FIELD OUTCOMES.

PROPOSED SOLUTION/ARCHITECTURE

Proposed Solution/Architecture

From Forecast to Field-Level Action — The 5-stage closed-loop pipeline.

1. PREDICT

Weather Telemetry

Ingest IMD/weather data to detect rainfall, heat, or moisture hazards.

2. ASSESS

Crop Phenology

Map hazard against specific crop variety and vulnerable growth stage.

3. PROTECT

Field Advisory

Send prescriptive action (e.g. drainage, mulch) in regional language.

4. OBSERVE

Field Photo

Farmer performs advisory and uploads a smartphone crop snapshot.

5. ADAPT

Stress Check

Compare before/after crop condition; update risk & next step.

THE CLOSED LOOP: Stage 5 field observations automatically feed back into Stage 1 continuous risk monitoring.

UNIQUE FEATURES/INNOVATION

Unique features/Innovation

The Continuous Feedback Loop — Why FarmShield is fundamentally different from a chatbot or weather app.

THE HERO INNOVATION

THE 5-STEP FEEDBACK LOOP

1. **PREDICT ➔ Micro-hazard risk identified 48h early**
2. **PROTECT ➔ Stage-specific intervention dispatched**
3. **OBSERVE ➔ Farmer uploads treated field photo**
4. **VERIFY ➔ AI checks if crop stress is recovering**
5. **ADAPT ➔ Recalibrate risk & trigger next action**

↺ **ADAPT loops back to PREDICT for continuous protection**

AI CROP STRESS VERIFICATION

Temporal Observation: BEFORE ➔ AFTER

STRESS IMPROVING
Leaf turgor recovering, water receded. Logs successful intervention protocol.

STRESS PERSISTING
Foliar stress stable; soil remains saturated. Dispatches secondary drainage advice.

STRESS INCREASING
Early root rot or pathogen emergence. Escalates to KVK Extension Officer.

★ **“FarmShield doesn't stop at an alert — it checks what happened in the field.”**

PROPOSED SOLUTION / FIELD WALKTHROUGH

Proposed Solution: Field Walkthrough

One Farmer. One Risk. One Intervention. — Real-world Maharashtra scenario.

📍 LOCATION: Satara, Maharashtra | 🌱 CROP: Green Pea | STAGE: Flowering (Sensitive to waterlogging)

STEP 01

🌧️ 70 mm Rain

Forecast Spike

STEP 02

🌱 Pea + Flowering

Vulnerable Stage

STEP 03

🔴 HIGH RISK

Hypoxia Warning

STEP 04

🛡️ Drainage Advisory

Dig Lateral Furrows

STEP 05

📸 Farmer Upload

Photo of Treated Field

STEP 06

🤖 AI Condition Check

Planned/Prototype CV

STEP 07

✅ STRESS IMPROVING

Flower Retention OK

🎯 **Projected Impact: Root stress + flower drop + potential yield loss.**

Timely drainage furrows prevented prolonged waterlogging, safeguarding flower retention during the critical node phase.

PROPOSED SOLUTION/ARCHITECTURE

Proposed Solution/Architecture: System I

Implemented ✓ | Prototype / Demo □ | Future Integration ●

Concept first, technology second — built for Indian rural connectivity realities.

1. DATA LAYER: Weather Telemetry • Satellite Data • Farmer Location

Tech: IMD & OpenWeather API ✓ • Soil Grids □ • Sentinel-2 NDVI ●

2. AI RISK ENGINE: Hazard Classification + Crop Stage Vulnerability Mapping

Tech: Crop-Stage Vulnerability Rules Matrix ✓ • Extreme Weather Thresholds ✓

3. ACTION ENGINE: Crop-Specific Prescriptive Intervention Advisory

Tech: ICAR / KVK Agronomic Protocols ✓ • Marathi / Hindi PWA ✓ • SMS/IVR Fallback ●

4. FIELD VERIFICATION: Farmer Photo Upload + Crop Condition Analysis

Tech: Mobile In-App Camera Upload ✓ • Planned / Prototype CV Model □

5. ADAPTATION LAYER: Temporal Delta Evaluation & Continuous Monitoring Loop

Tech: Temporal Trend Engine □ • Closed-Loop Feedback System ✓

Architectural Principle: Core advisory functions offline; image analysis runs when connectivity is available.

EXPECTED OUTCOME

Expected Outcome

From Reactive Farming to Early Intervention — Impact on Indian smallholders.

SYSTEMIC SHIFT: REACT AFTER DAMAGE ➔ ➔ ➔ PREDICT + ACT + VERIFY

EARLIER WARNING

Detect multi-hazard climate risk 48–72h prior to severe damage, providing smallholders with a vital intervention window.

SMARTER WATER RESPONSE

Actionable drainage and moisture protocols mitigate both flood hypoxia and dry-spell stress.

CROP-SPECIFIC GUIDANCE

Tailored advice based on crop variety and growth stage prevents wasted inputs and ineffective chemical spraying.

CONTINUOUS GROUND EVIDENCE

Farmer-uploaded image tracking gives smallholders confidence and extension workers (KVK) verifiable field proof.

Designed specifically for Indian smallholder farming contexts, low-connectivity zones, and regional languages.

CONCLUSION & VISION

Protect the Crop Before the Loss.

PREDICT • PROTECT • VERIFY • ADAPT

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